

KTH ROYAL INSTITUTE
OF TECHNOLOGY

Degree Project in the Field of Technology and the Main Field of Study Computer Science and Engineering

Second cycle, 30 credits

Between Water and Air: An Exploration of Somaesthetic Drone Interaction Informed by the Embodied Experience of Freediving

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Stockholm, Sweden, 2025

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Master's Programme, Interactive Media Technology, 120 credits
Date: September 6, 2025

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Swedish title: DMellan vatten och luft: En utforskning av somaestetisk drönarinteraktion informerad av fridykningens förkroppsligade upplevelse

Between Water and Air: An Exploration of Somaesthetic Drone Interaction Informed by the Embodied Experience of Freediving

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This thesis investigates how movement qualities inspired by freediving can inform the design of somaesthetic human–drone interaction. Using a first-person Research-through-Design (RtD) methodology, the process began with a micro-phenomenological interview to extract nuanced somatic qualities in my past freediving experience, which were translated into the experiential quality of *force projection*. Two key drone interaction patterns, *Follow like Fins* and *Wave from Pushing*, were developed to express this quality. These were further enriched through aesthetic augmentations that treated the drone's airflow as a design material. The outcomes and insights generated through this iterative process were synthesized into two speculative scenarios that reimagine drones as expressive partners in performance and play. This thesis contributes to the emerging area of somaesthetic drone interaction by offering a reflective account of the entangled relationship between soma, material, and technology.

SAMMANFATTNING

Denna avhandling undersöker hur förkroppsligade rörelsekvaliteter inspirerade av fridykning kan informera utformningen av somaestetisk människa–drönare-interaktion. Genom att använda en förstapersonsbaserad Research-through-Design (RtD) metodologi inleddes processen med en mikrofenomenologisk intervju för att extrahera nyanserade somatiska kvaliteter från min tidigare fridykningsupplevelse, vilka översattes till den upplevelsebaserade kvalitén *kraftprojektion*. Två centrala interaktionsmönster med drönare, F*ölja som fenor* och *Våg från tryck*, utvecklades för att gestalta denna kvalitet. Dessa berikades ytterligare genom estetiska förstärkningar där drönarens luftflöde behandlades som ett designmaterial. Resultaten och insikterna som genererades genom denna iterativa process syntetiserades i två spekulativa scenarier som omformulerar drönare som expressiva partners i scenkonst och lek. Denna avhandling bidrar till det framväxande området somestetisk drönarinteraktion genom att erbjuda en reflekterande redogörelse för det sammanflätade förhållandet mellan kropp, material och teknologi.

CCS Concepts: • **Human-centered computing** → **Interaction design**; *Interaction design process and methods*.

Keywords: soma design, drones, human-drone interaction, freediving

Nyckelord: soma design, drönare, människa–drönare-interaktion, fridykning

1 INTRODUCTION

In recent years, somaesthetic design has emerged as a design practice within the field of Human-Computer Interaction (HCI), focusing on designing from and for the lived, sentient body, the soma. This perspective foregrounds bodily awareness, inward attention, and sensory engagement as central to the design process [13]. While somaesthetic principles have been applied to interaction design [23], intimate technologies [38], and performative arts [17], their potential also extends to the domain of Human-Drone Interaction (HDI). Drones, often framed as mechanical tools for navigation or surveillance that prioritize functionality and performance in the past [43], are increasingly being explored

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Document date: September 6, 2025

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as expressive agents. This shift opens up new possibilities for somaesthetic design by introducing the presence of an autonomous, responsive agent [2] that can not only act as a medium that encompass somaesthetic experience but also a relational partner in movement and bodily expression.

In particular, La Delfa et al. demonstrated how movement-based experiences such as Tai Chi can be used as design metaphors that inform somatic drone interaction [19]. Drawing inspiration from their work, this thesis introduces my past experience of freediving as a source of design ideation. In this personal experience, freediving provided an experience that aligns closely with somaesthetic values. It involves a heightened inward attention to bodily signals such as the urge to breathe, shifts in ear pressure, and the alternating tension and relaxation of muscles, all taking place in a deeply focused and calm mental state. The connection between somaesthetic drone interaction and freediving is grounded in this shared emphasis on bodily awareness, internal sensation, and expressive movement, making freediving a fertile ground for imagining new forms of movement-based drone interaction attuned to somatic experience.

The aim of this thesis is to explore the design space of somaesthetic human-drone interaction through incorporating movement qualities drawn from freediving. The transition from underwater experience into land-based interaction follows a unique design trajectory, holding a potential to uncover innovative forms of interaction experience between humans and drones. This leads to the research question:

How can a personal freediving experience be translated into somaesthetic drone interactions through a Research-through-Design process?

Since this is a qualitative and explorative thesis, the research question is intentionally open-ended to encompass a broader range of insights and findings. This research adopts a Research-through-Design (RtD) methodology [47] through a first-person perspective [14] in a somaesthetic worldview [32], that design decisions are arrived at through both experiential knowledge and the embodied, subjective feelings encountered during the exploration process. The primary contribution of this work lies in the knowledge generated through the design exploration, which unfolds through iterative cycles of bodystorming, prototyping, and material experimentation. Experiential qualities were first extracted from a freediving experience using micro-phenomenological interviews. These were then translated into movement-based drone interactions by ideating with corporeal bodies—both my own and the drone's. Finally, two speculative scenarios were developed to synthesize and express the insights gathered throughout the design process. These processes showcased the continual translation and transition between concepts and realizations, mediums and materials, implying a dynamic and reflective design journey that gradually constructed the drones into expressive agents of embodied interaction.

The contribution of this thesis is threefold:

(1) A first-person RtD process of how movement qualities from a personal somatic practice can inform somaesthetic drone interaction design;
(2) A set of design outcomes, including two drone interaction metaphors, two drone-augmented aesthetic experiences, and two speculative scenarios that synthesize design insights, and;
(3) A reflective account that exemplifies how meaning, conceptually, materially, and somatically, emerged through the iterative design process.

2 LITERATURE REVIEW

Here, I present related literature on (a) somaesthetic design and experience, and (b) human-drone and somaesthetic drone interaction.

2.1 Somaesthetic Design and Experience

Somaesthetics, a term that combines ‘soma’ and ‘aesthetics’, was introduced by the philosopher Richard Shusterman [35]. The disciplinary field emphasizes the cultivation of sensory appreciation through experiencing of our soma — the sentient, perceptive, and living body. Rather than viewing the body, without the mind, as a mere physical entity, somaesthetics recognizes it as an active, feeling, and responsive medium integral to human experience and interaction [13].

In the fields of HCI and interaction design, somaesthetic approaches and principles have been used to develop both design methodologies and artifacts. Höök introduced the strong concept of somaesthetic appreciation, which outlined four essential qualities: subtleness, making space, intimate correspondence, and articulating experience, providing a foundational framework for soma design [15]. Unlike usability-driven metrics that are important factors throughout conventional design iteration, somaesthetic design prioritizes engaging the whole body, increasing sensory awareness, and embracing embodied sensory experience. In this approach, no experience is considered undesirable - whether it involves unfamiliar sensation [23] or even pain or discomfort [29, 44] - as all sensations contribute to deepening our understanding of the soma and its interaction with the world. Therefore, it leads to the core principle of somaesthetics, by realizing and embracing this understanding, we can enrich our lives and experience of the world. [38].

In soma design, experience is understood as a phenomenon shaped by its specific context, emphasizing its inherent uniqueness, subjectivity, and dynamic, multi-dimensional nature [31, 40]. To capture, articulate and analyze these rich experiential qualities, first-person methods, such as autoethnography and autobiographical design, can offer detailed, expressive, and direct means of access [14, 27]. First-person methods can be applied in different stages of design iteration. They can serve as important exploratory tools that allow designers to immerse themselves in the design space. For example, Höök derived a design quality named ‘rhythm’ through her horseback riding experience [12]. Additionally, they could be used in the iterative design process, exploring and informing potential design decisions made throughout the process [27]. Another approach is to extract experiential qualities through articulation methods such as micro-phenomenological interviews. Unlike regular interviews, which mostly obtain general descriptions and explored causal relationships of experiences, incidents or movements, micro-phenomenological interviews are a technique that elicits detailed descriptions of an evoked experience [31]. They enable individuals to articulate the rich details across multiple dimensions of an experience, regardless of whether it is adopted through a self-experienced first-person perspective or second-person dialogue.

While there is no definitive procedure for designing somaesthetic bodily experiences, embodied sketching [24] serves as a key methodology to keep the designer closely attuned to the soma, evoking first-person sensory experiences throughout the design process. Sensitizing a designer’s soma enables a deeper exploration of the design space by cultivating heightened bodily awareness and uncovering new movement possibilities [13]. Common sensitization concepts include change and interest and estrangement [13, 14]. The former involves subdividing attention to specific bodily areas, fostering a more granular awareness of sensation and movement through methods such as Feldenkrais-exercises [39]. The latter disrupts habitual behaviors, introducing unfamiliar movement patterns to encourage new ways of sensing and engaging with the soma [23, 42, 46]. Both techniques engage the soma to highlight subtle nuances and variations in bodily experience, expanding the range of possible interactions in design exploration. Additionally, bodystorming provides a design ideation method grounded in subjective bodily feeling, where designers physically enact interactions to gain embodied insights into movement, posture, and affect. By directly experiencing movement-based interactions, bodystorming facilitates the generation of body-centered design ideas, reinforcing the somaesthetic approach to interaction design [24, 34].

2.2 Human-Drone and Somaesthetic Drone Interaction

Drones are autonomous or semi-autonomous aerial systems capable of navigating environments. Their unique functionality and materiality grant them great potential for integration across a variety of domains—whether practical, experimental, or artistic. This evolution led to a change in how drones are designed, perceived, and evaluated: moving beyond mere technical tools to become socialized agents [43]. Baytaş et al. coined the term 'social drones', referring to drones that are fully autonomous and operate in spaces populated by human users or bystanders [2]. The interactive nature of socialized agents raised critical questions about the relationship between humans and drones, prompting researchers to draw on perspectives and methodologies from the field of HCI.

The dominant perspective in HDI approaches the experience of social drones from the user-oriented perspective. This includes investigations into how to integrate drones into different scenarios and in these scenarios how users perceive the role and behavior of the drone [2, 4, 16], how to increase efficiency and effectiveness through intuitive control and communication of drones [26], and the possible emotional attachment [10] that could be developed. These research insights can inform design decisions that address human-centered and drone-centered concerns [11]. Cauchard et al. for example reported on the natural adaptation of interaction models from interacting with people or pets to drones, suggesting an intuitive gesture control derived from existing relationships [4].

In my thesis, I examine drones from a different perspective, challenging the user-centric perspective by positioning them as active participants in dynamic human-machine relationships, rather than merely passive objects in interaction [45]. This approach embraces the inherent complexity of interactions between human and technology and their potential for novel embodied experiences [41]. Instead of asking 'how can drone interaction be intuitive?', these studies shift the focus to 'what could drone interaction feel like?', exploring the nuances of drone behavior and interaction dynamics. Here, I draw inspiration from another strand of research that has introduced somaesthetic perspectives within HDI, emphasizing the role of bodily experience and sensory engagement in human-drone interaction.

The exploration of somaesthetic qualities in human-drone interaction utilized drones as the medium to explore new forms of interaction that are deeply tied to somatic awareness and bodily movement. Autonomous drones, capable of independently responding to bodily signals or movements [7, 8, 19, 20, 36], open up new possibilities for appreciating and engaging with bodily aesthetics in drone interaction. Additionally, the concept of mechanical sympathy extended somatic interaction from feeling the sentient body to the mutual somatic relationship between machine and human, perceiving human and the drone as interactive dyad that built upon intercorporeality among agents [20]. With participants adjusting their movement according to the drone's ability and limitation, and establishing the aesthetic connection with it, the drone became the collaborator of discovering bodily potential [8], co-shaper of the interaction [20, 36], and the co-creator of an artistic performance [7]. In particular, La Delfa et al. integrated the movement qualities of Tai Chi within drone interaction, which showcased how real world bodily movement could inspire somatic qualities being developed in drone interaction [19].

3 METHOD AND DESIGN PROCESS

Here, I present (a) an overview of my method and design process, and detail descriptions of the design explorations: (b) the extraction of experiential qualities, (c) the exploration of design space, (d) the ideation with drones, and (e) the development of aesthetic experiences.

3.1 Method Overview

This thesis adopts a Research-through-Design (RtD) approach [47] with the aim of translating movement qualities from freediving into design metaphors that inspire novel forms of human-drone interaction. This translation—from

movement to high-level design ideas and into concrete design language (materials, aesthetics, technologies, and interaction patterns)—is carried out through an iterative process involving a series of active design explorations and choices. In design research, it is precisely through the rationale behind such decisions that both the processual knowledge and the final artefacts are considered valid contributions to knowledge [47].

The design process unfolded in four distinct phases: (1) extracting experiential qualities from my freediving experience through a micro-phenomenological interview, (2) exploring the design space of these qualities using bodystorming and sketching, (3) ideating drone interaction patterns grounded in the core quality through the iterative process of bodystorming with drones, and (4) constructing an integrated human-drone interactive experience with explorations on physical prototype. Furthermore, all phases relied on first-person methods to examine the somaesthetic dimensions throughout the design process [12, 14, 27], as such approaches offer detailed and expressive means to capture the nuances, the subjective and embodied experience that arise during soma design process.

3.2 Extracting Experiential Qualities – Micro-phenomenological Interview

In soma design, the exploration of movement qualities is usually done through practical engagement with the embodied experience, and then articulating the feelings and sensations perceived by the sentient body. However, there are environmental constraints of performing freediving in Sweden, for instance, temperature and location. Therefore, I instead participated as a subject of a micro-phenomenological (M-P) interview to recall the experiential qualities of my freediving experience through elicitation. As described by Prpa et al., a M-P interview is a guided method that supports participants in accessing the subtle, embodied dimensions of lived experience, often reaching layers that extend beyond surface-level descriptions. This includes the synchronic aspects, implying the cognitive, sensory, and emotional details that exist at one point in time, and the diachronic ones, which refer to the unfolding of experience over time. M-P methodology is particularly effective in capturing tacit knowledge and articulating qualities into vocabularies that are suitable for experience design [31]. This interview was conducted by Katerina Stepanova, who is a certified facilitator of the M-P interview, and took approximately one hour, facilitating the detailed recollection of embodied movement and the evocative sensation from a previous freediving experience six months ago.

The richness and multi-dimensionality of the experience unfolded not only during the interview but also before and after it. Prior to the interview, I reflected on the standard procedures of freediving and began to recall specific descriptions of my experience, focusing particularly on what actions I performed at each stage of the procedure, what events unfolded, what feelings arose, and how I responded to various scenarios throughout the dive. This preparatory reflection laid the foundation for a deeper exploration during the interview.

During the interview, the facilitator used questions beginning with ‘how’, prompting me to delve further into layers of experience. These prompts also supported me in articulating inward somatic sensations and tacit knowledge, often expressed through metaphors or vocabulary drawn from the external world. For instance, I described the feeling of staring at the sand flowing in front of me as ‘like being exposed in a visual version of white noise’. This mapping between feeling and metaphors implies an articulation on a subjective and elusive embodied experience into languages that allow for clear understanding and communication [20, 31].

After the interview, the audio recording of the interview was revisited to capture notable quotes and embodied feelings, forming the basis for an autobiographical account of the experience. While writing, the process was further enriched by revisiting the video recording of my previous dives and integrating first-person micro-phenomenological prompting techniques, enabling the recall of nuances and subtleties of the experience. The completed autobiography was then reviewed to extract meaningful experiential qualities: *from alien to extension of body, unfamiliar movement and space, adjusting techniques by observing our limitation, transporting between two worlds*, *and balancing inward and outward attention*.

From these, my focus narrowed to one quality which strongly resonate with drone interaction. *Alien to extension of the body* describes how the fins, initially perceived as a foreign attachment, gradually became an integrated part of my body through a process of mutual adjustment and attunement over time. I was drawn to this quality because the relationship between diver and fins, reminded me of sensations that I had previously experienced when I engaged with the How To Train Your Drone interaction [21], where the drone responded to my movements in a way that made it feel like an externally integrated part of my body. However, rather than serving as a generative metaphor, this quality more closely resembles a resulting sensation of an interaction. To unpack this further, I revisited the corresponding autobiographical passages. In the M-P interview, I had remarked: *"While kicking fins, the hips is swinging like dancing-left and right, left and right. This movement of the glutes propagates through the legs, into the feet, and gradually into the fins."* This description highlighted how the fins, as an extension, respond to bodily force and transmit energy outward in a laggy behavior. From this, the concept of force projection began to take shape, becoming a focal point for the next iteration of exploration: how force projection could be expressed, extended, and made perceivable through bodily and drone interactions.

3.3 Exploring Design Space of Force Projection and Design Metaphors

To explore the *force projection* concept, I conducted bodystorming sessions to investigate how bodily movements generate and project force into space. In the first session, various actions were experimented with, including pushing, pulling, pushing aside, pressing, lifting, jumping, kicking, and stepping. These movements were performed with a scarf tied to the body, serving to visually and physically amplify the projection and extension of force through space (Fig. 1a). The entire session was recorded and immediately reviewed to produce a force sketch, a visual reflection aimed at capturing and analyzing the dynamics and flow of bodily forces expressed during the exploration (Fig. 1b).

The force sketch illustrated which muscles intensified during specific actions (marked in pink), the direction and type of force generated (marked with red arrows) and how the force diffused and radiated outward into space (marked with blue arrows). This session mapped a range of basic bodily movements associated with force, forming a foundation for later interaction design. Furthermore, the sketches emphasized that force not only acts along a direct trajectory but also behaves like a wave, diffusing spatially beyond the line of action. These insights will serve as inspiration in the subsequent stages of the design process.

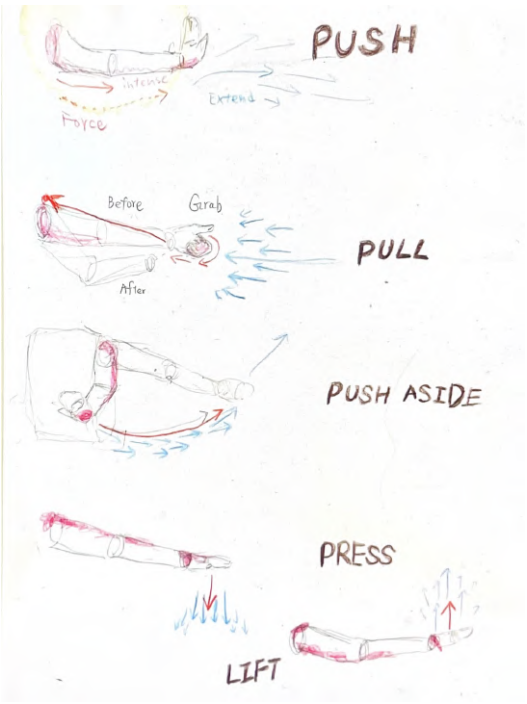
3.4 Ideation with Drones - Exploration with Drone's Movement and Interaction

Further bodystorming sessions involved exploring bodily interactions with the Crazyflie[1], educational-purpose nano-drones that can be extended with various hardware decks to enable different functionalities. In these sessions, the goal was to investigate how a drone might reflect or express the force exerted on it through its movement modalities, such as position, speed, direction, tilt angle, or even other sensory modalities. However, recognizing that technologies come with inherent capabilities and limitations and that Crazyflies in particular are not consumer-grade drones, they do not behave as a smooth, seamless interactive agent and are prone-to-error from a user experience perspective. It is important to note that these limitations largely stem from the fact that Crazyflies are programmable and open-source platforms. Unlike commercial drones that offer polished experience, Crazyflies demand substantial technical input from the user to achieve similar levels of reliability and performance. Therefore, in these sessions, alongside bodily experimentation, attention was also directed toward understanding the drone's constraints. From a broader perspective, these limitations can be reframed as unique behavioral expressions. For instance, the involuntary wiggling behavior of the drone as it attempted to stabilize itself while falling was repurposed as a channel to express affective qualities such

[1] See official website: https://www.bitcraze.io/products/crazyflie-2-1-plus/



(a) Bodystorming session using scarf to visualize the force flow while I'm performing *pull*.



(b) Force sketch derived from bodystorming, expressing the dynamics the movement.

Figure 1: Exploration of design space of force projection

as desperation [7]. As such, observing and engaging with these constraints proved valuable not only for identifying technical boundaries but also for uncovering opportunities to design distinctive interactions that could meaningfully incorporate and respond to the drone's specific characteristics.

In one session, *push*, originally derived from the earlier force sketch, was selected for implementation in the drone interaction. This movement was chosen for its correspondence between bodily action (pushing) and the drone's responsive behavior (being pushed away), as well as its feasibility for implementation. Drones used in this session were equipped with a Multi-ranger deck[2] and a Flow V2 deck[3], enabling them to measure distance in four directions and support stable position-hold via local positioning flight control. The aim was to explore whether a metaphorical push interaction, applied through hand movement, could evoke an embodied sense or imaginative perception of force through the drone's responsive feedback. In addition, to document and reflect on the sessions, sketches were used extensively. These served as a means of recording the design process and as tools for ideation, analysis, and guiding design decisions in every exploration iteration.

To simulate push, the drone was configured to move away a fixed distance after detecting an obstacle below the distance threshold. Several pairs of parameters were tested, for example, a hand–drone distance below 40 cm would trigger the drone to move away by 40 cm. However, all these trials did not evoke a convincing sensation of pushing force between the hand and the drone. Instead, it felt more akin to the repelling effect of two similar magnetic poles.

[2]See product page: https://www.bitcraze.io/products/multi-ranger-deck/
[3]See product page: https://www.bitcraze.io/products/flow-deck-v2/

The embodied perception of force typically involves varying intensities of muscular engagement depending on the amount of force exerted. Moreover, different levels of strength usually elicit distinct responsive behaviors from the object being acted upon. However, in this scenario, the drone's backward movement was based solely on proximity sensing and a fixed retreat distance, rather than responding dynamically to the quality or intensity of the detected movement. As a result, the interaction lacked the somatic richness necessary to simulate a convincing sense of push. As Norman argued, gestural or movement systems often fail when feedback that allows users to understand and control their actions [28]. Without dynamic, interpretable feedback tied to the nuance of bodily gestures, the system risks becoming a mechanical reaction rather than a meaningful interaction.

To address this, adjustments were made to enhance the perceived strength and force within the interaction. Specifically, the drone was reconfigured so that faster hand movements would cause it to move backward at higher speeds, and closer hand proximity would result in a greater retreat distance. These refinements aimed to establish a more dynamic and embodied interaction loop with the drone. The interaction offered greater responsiveness between actions and drone behavior, and was perceived as an engaging and playful way to control the drone, according to a peer in an informal trial session. Moreover, this interaction resembled a similar interactive method as employed in previous work on the Drone Arena [36]. However, from a somaesthetic perspective, it lacked a critical quality: intimate correspondence [15]. This term refers to an immediacy and synchronization between system feedback and rhythms and flows of the body—where the technology and the soma come into a shared rhythm or understanding.

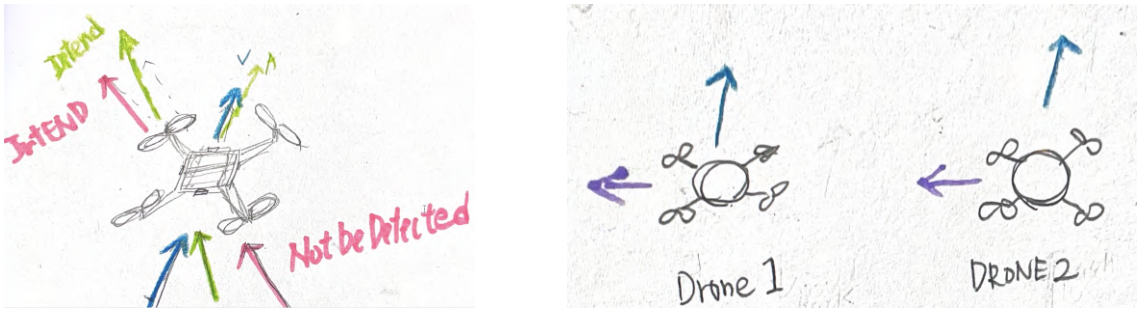
Due to limitations in the sensitivity of distance sensors, the feedback was often imprecise or inconsistent. For example, the same approaching velocity or hand distance could result in different outcomes across trials. This lack of consistency made it difficult for me to establish a meaningful bond with the drone, where it emerges through mutual responsiveness and interpretability. Such a bond depends on developing a shared 'language' between human and system. But how can one learn a language when the mapping between words and meaning change unpredictably? Through this initial exploration, it became obvious that the interaction needed to be more delicate to support somatic engagement. This realization led to the development of two interaction methods with improved delicacy, referring to a greater expressiveness or responsiveness between drone's movement and bodily cues. Both interactions were derived from the original *push* concept, and each linked to specific metaphors derived from freediving.

3.4.1 Metaphor 1 - Follow like Fins. One potential approach to increase the delicacy of interaction was to consider fine-grained detection and movement. The force sketch had already emphasized how force is projected into three-dimensional space, not just in terms of magnitude but also orientation. However, this 'direction of force' was a component that was overlooked in earlier exploration iterations with drones. This raised the question: How could drone interaction reflect such directional force and how could this be technically realized?

In earlier explorations using the Multi-ranger deck, detecting directional input proved difficult. Specifically, the deck is equipped with four distance sensors positioned along the drone's cardinal axes. This configuration makes it impossible to distinguish between an oblique approach and a direct perpendicular approach, as both are interpreted similarly. Additionally, approaches aligned with the orientation of the propellers fall into a sensor blind spot, meaning that the system is unable to register movement from those directions (Fig. 2a). These limitations significantly constrained the ability to reflect nuanced directional forces in the interaction.

In contrast, using the Lighthouse deck[4] in conjunction with Lighthouse base stations enabled global spatial tracking of the drone's position with greater stability and precision. This opened up the potential for designing interactions that respond to the direction of movement within a mapped space. One simple yet effective realization of involving the

[4]See product page: https://www.bitcraze.io/products/lighthouse-positioning-deck/



(a) Limit on direction detection using Multi-ranger deck, where certain directon of approaching couldn't be differentiated or even detected.

(b) Directive drone movement using the Lighthouse deck, where the fine-grained positional changes of one drone could be detected and pass to the other.

Figure 2: Exploration of directive drone movement in response to the direction of force

direction is using one drone as a sensor and another as a responsive agent (Fig. 2b, Fig. 4a). The sensor drone, held by hand, captures the movement input, while the responsive drone mirrors or projects that movement in space.

This interaction strongly reminded me of the leg movements while wearing fins underwater. This connection became particularly vivid when introducing a slight delay between input and response in the drone interaction. To be specific, when moving the hand-held sensor up and down, the drone that act as the responsive agent, might still be reaching its highest point even as the hand had already begun pulling downward (Fig. 3). In freediving, fins are typically long and flexible, with large surface areas and pliable materials that make them highly responsive to water resistance. However, their shape and material properties introduce a temporal lag between the diver's movement and the corresponding motion of the fins. The energy from a kick travels gradually through the fin, causing each part of it to gradually bend and follow. This delay creates a rhythm, a temporal gap, in which the diver must wait for the force to transfer through the medium before the fins complete the motion. Freedivers learn to synchronize with this rhythm, becoming attuned to the release of water resistance and developing an awareness of the subtle feedback that emerges between soma, medium, and extension.

Similarly, when the drone followed my movement with a slight delay, it evoked the same embodied rhythm experienced in freediving. This delayed response established a loop of interpretation and responding, where I gradually developed a somatic sense of the appropriate tempo and speed needed for the drone to respond fluidly, as if one were waiting for it to catch up. This adjustment fostered a subtle but meaningful connection, where control is not enforced but negotiated through somatic perception and responsive awareness.

3.4.2 Metaphor 2 - Wave from Pushing. Another attempt to increase the delicacy of the interaction involved shifting attention away from precise hand-to-drone mapping, and instead focusing on generating smooth, expressive drone movements. Upon analyzing footage from my past freediving sessions, I observed that the shape and dynamics of the fins closely resembled the one of a wave. This observation gave rise to the design metaphor *wave*. Waves, whether in the ocean, in sound, or in other physical phenomena, are forms of energy that propagate through a medium over time. A force, when applied to a medium, creates ripples or oscillations that extend outward. For instance, dropping a pebble into water generates concentric ripples from the point of contact; sound is produced by vibrating air particles that transmit energy through space.

In this metaphor, the body of water experienced in freediving is reimagined as a body of air through which drones move. Just as divers navigate through water, drones float and maneuver through air, also responding to external forces

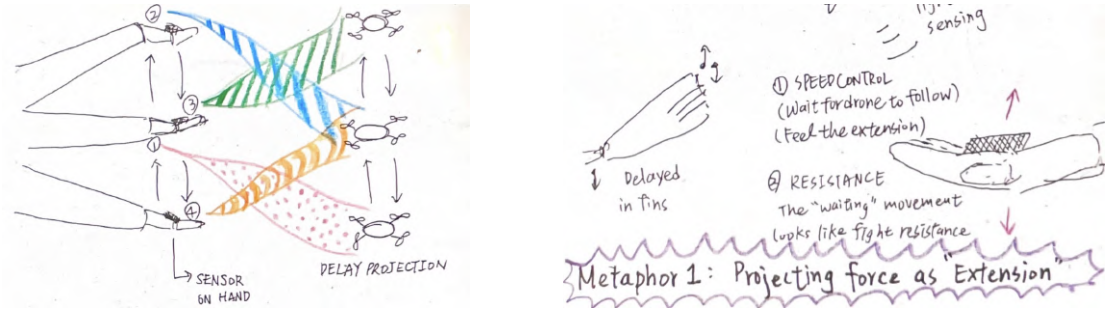


Figure 3: The similar embodied qualities (delayed response) and somaesthetic feeling (pace controling) between the drone interaction and fin-kicking movement

or environmental changes. Within this context, a bodily action, such as a push, can be seen as a disturbance to this medium, generating wave-like movements. The drone, as an object within this medium, thus becomes a vessel through which the influence of disturbance is, externalized and made visible. Inspired by this, I implemented a push to backward movement using a damped sine wave (a sine wave with gradually decreasing amplitude) via the Flow V2 deck. The aim was to explore whether such a wave pattern, intending to simulate the ripple effect generated by disturbing a water surface, could be an evocative motion following a triggering push (Fig. 4b).

During the exploration, several variations of drone movement were tested, including a normal sine wave and integration with spiral patterns. However, these experiments and the observation suggested that overly complex movement patterns may compromise somatic qualities. When movement becomes complex, it disrupted my ability to attune to the system's feedback, weakening the embodied dialogue with the interactive agent. In contrast, clarity in movement and responsiveness was essential to maintaining a meaningful and interpretable interaction loop. This observation also aligns with recommendation that close-range human-drone interaction is most suitable for simple control mappings [19]. In this regard, the damped sine wave emerged as an ideal movement pattern as it achieves the expression of natural energy propagation without adding unnecessary complexity to the interaction.

While the recorded damped sine drone movement visually revealed a beautiful, fluid motion, evoking the sense of waves propagating through space, the wave pattern was less perceptible in real-time, first-person interaction. This was primarily because the backward movement occurred along the same axis as my line of sight, making it difficult to perceive. In contrast, vertical (up–down) motion along the z-axis remained more visually salient. Therefore, to effectively convey the flow and motion, it was necessary to augment or externalize the presence of the wave beyond drone movement alone—through other sensory cues that amplify the perception of the aesthetic experience of a wave.

3.5 Developing Aesthetic Experience of Waves

A key requirement while ideating the aesthetic experience of the wave was that it should be integrated into existing drone interaction, not merely an external or decorative element. In the course of this process, I became intrigued by the air displacement naturally produced by the drone during flight. This inspired the idea of treating the drone's physical characteristics as a design material. This phase of exploration focused on understanding the characteristics of the airflow generated by the drone, observing its effects, and investigating how it could be harnessed to create wave-like effects.

To begin with, I placed various lightweight materials underneath the hovering drones to visualize the airflow. One particularly informative test used styrofoam packaging (Fig. 5a), which I used to create a visualization akin to a wind field map. Specifically, a strong downward air current was observed directly beneath the drone. Upon reaching the



(a) *Follow like fins* interaction, where the flying drone mirrors the movement of sensor drone.



(b) *Wave from pusing* interaction, where drone move backward in a wave-shape triggered by push

Figure 4: Bodystorming with drones, exploring two metaphors: *Follow like Fins* and *Wave from Pushing*



(a) When placing styrofoam under the flying drone, it visualize the strong downward airflow and the dispersed side airflow generated by the drone.

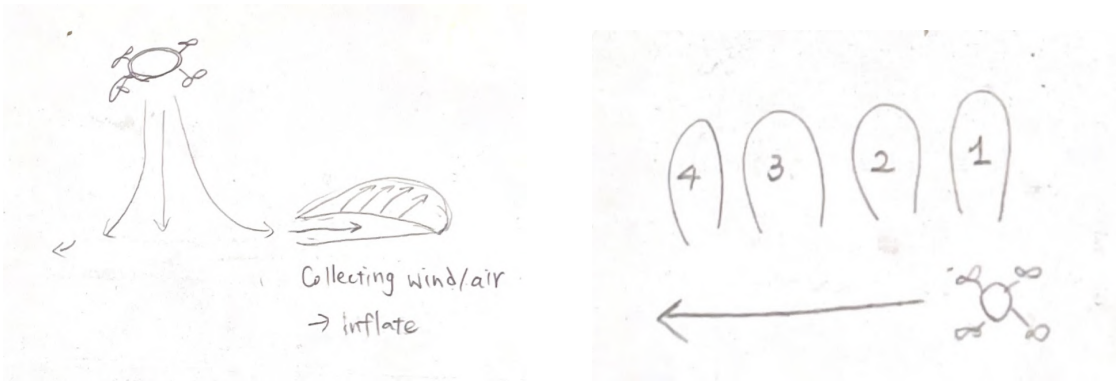


(b) When placing the flying drone next to the cloth, the airflow of the drone will enter through its side and create inflation.

Figure 5: The use of styrofoam and cloth to visualize air displacement generated by drones

floor, this current dispersed outward to the sides, creating noticeable lateral air movement along the ground. Another material, fabric, demonstrated that when air entered from the sides, it could be collected, resulting in inflation (Fig. 5b). I developed two low-fidelity conceptual prototypes from these observations.

3.5.1 Concept 1 - Air Collector, Inflate. The first concept was to create a wind collector that inflates as a drone passes nearby and deflates when it moves away (Fig. 6a). This wind collector consisted of a covering that is closed on one side, capturing the airflow to produce the inflation effect, and a supporting frame that maintained the overall shape and kept the entrance open to allow air to enter while in a deflated state. When several collectors are arranged



(a) An installation that collect the airflow **(b) Collectors resemble wave while drones passed by**

Figure 6: Concept of utilizing air displacement of drone for inflation to simulate wave effect

in sequence, each responding at different times or inflating at slightly different locations, the installation creates a wave-like effect (Fig. 6b), similar to the 'human wave' often seen in stadium crowds.

This concept was first explored using plastic bags as air collectors. Through several trials, several key design parameters were identified, including the geometry of the frame of holding the bag, and the texture of the covering material, both of which influenced the visual and dynamic quality of the simulated wave. First of all, chicken wire was used to craft the frame, enabling the creation of irregular upward and downward curves (Fig. 7a). These curves allowed the covering to form a visually appealing wave-like shape when in the deflated state.

As for the covering material, it needed to be lightweight enough to respond to airflow, yet also capable of resetting its form between inflations. Plastic bags are visually expressive, especially when light reflects off their wrinkled surface in a way that resembles glimmers on the ocean (Fig. 7b). However, they are overly sensitive to interference. Even slight air displacement from drones in nearby areas could trigger unintended inflation, disrupting the intended wave effect. Moreover, once inflated, plastic bags struggle to return to their original form attaching to the frame, often remaining semi-inflated and thereby interfering with the subsequent inflation cycle. Light fabric, on the other hand, offered a more controlled behavior. Slightly heavier than plastic bags, therefore it was less affected by minor currents and would reattach to the frame after deflation (Fig. 7c). This allowed for rhythmic cycles of inflation and deflation, which in turn enhanced the perceptibility and integrity of the wave interaction.

3.5.2 Concept 2 - Shimmering Panel, Flutter. Another idea emerged during a bodystorming session with the supervision team: the fluttering effect of shimmering wall panels. Commonly used in celebratory settings such as parties and weddings, shimmering panels are made of lightweight reflective pieces that flutter in response to vibration or air movement. These commercial shimmering panels are designed with small pillars or pegs that suspend each piece slightly above the base surface. This structure allows air to circulate underneath and provides just enough freedom for the pieces to sway when disturbed by external forces. The flutter combined with lighting could produce vibrant, fragmented reflections that simulate the glimmering of the water surface. This resembled the sparkling patterns created when sunlight penetrates water and scatters across its undulating surface. The exploration of shimmering panels focused on two primary aspects: (1) the implementation of fluttering effects utilizing drone-induced airflow, and (2) the lighting and projection requirements to construct the visual effect.

Firstly, the placement of the shimmering panels would significantly influence the ease and extent of the fluttering effect. Two effective configurations were identified through experimentation. In the first, the panels were laid flat on



(a) chicken wire wave-shape frame **(b) Plastic bag as covering** **(c) Light cloth as covering**

Figure 7: Chicken-wire wave-shape frame along with different coverings

the ground, forming a horizontal shimmering surface. When a drone hovered directly above, the downward airflow triggered a strong fluttering effect (Fig. 8). In the second configuration, the panels were arranged vertically to create a shimmering wall. In this case, the fluttering was activated by lateral airflow, generated when the downward thrust hit the floor and spread outward. However, this side current is weaker than the direct downward flow, making the placement and distance between the drone and the panel much more constrained.

Secondly, the placement and selection of the light source, as well as the installation of the projection plane all had a significant impact on the visibility of the shimmering effect. Ideally, the light source should have a narrow beam angle and be positioned close to the panels. To my surprise, high lumen output was not strictly necessary. These are concluded based on tests on different light sources, including mobile phone flashlights, single LED bulbs, and LED strips. It was found that mobile phone flashlights, despite their relatively low brightness, produced the best results. This is likely due to their focused beam angle, which is well-suited to interacting with the semi-reflective surfaces of the shimmering pieces. In contrast, LED strips provided diffuse lighting that failed to create the desired projection effect in regards to their high lumens.

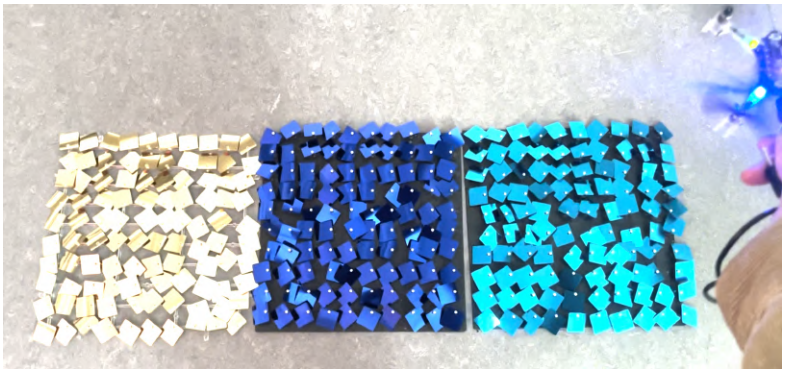


Figure 8: Flutter of shimmering panel while drone flying on top

It is important to note that these conditions represent an optimal setup based on available resources. The requirements would likely shift with the use of alternative or more advanced materials. After all, prototyping is inherently about resource management: understanding what is available, how to use it, and how to optimize limited resources to achieve intended outcomes.

4 SPECULATIVE SCENARIO

Through the design exploration, I successfully integrated the metaphor of force projection into drone interaction. Several experimental concepts, such as the use of air displacement to trigger inflation and shimmering effects, were also demonstrated to be feasible. To envision these concepts within a more immersive experience, I chose to shift from physical prototyping toward storytelling, which allow for greater freedom in exploring the potentials and values of the insights. According to Blythe, narratives in RtD could establish a discursive space where the concepts remain a degree of ambiguity for discussion [3]. Therefore, in the following section, I present two speculative scenarios using a storytelling approach along with concept illustration which emphasize on the overall atmosphere. These narratives envision possible outcomes that build upon the insights from the design process, illustrating the potential of movement-inspired somaesthetic drone interactions.

4.1 Artistic Performance

This scenario was inspired by Dancing with Drones [7] and Aeroquake [18], where they revealed drones could become a complementary artistic expression of human bodily movement. Within this context, the aesthetics of modern dance—characterized by the fluid dynamics of outward, expansive gestures and spatial flow—naturally lend themselves to guiding the drone across the stage. The action aligns with the movement within *Follow like Fins* interaction, integrated with the inflatables concept, could turn the drone into a graceful companion within the dance performance (Fig. 9):

> *The performance begins in a small theatre, with the audience seated all around the stage. As the ambient lights dim, the stage lights brighten, revealing a floor covered layered with sheer fabrics. Some lie flat, others curl slightly, forming soft hills and valleys like a quiet, sleeping sea.*
> *Soft, fluid music begins. A dancer enters, weaving gracefully between the fabric landscape, his body flowing as if carried by an unseen current. His movements seem as though rehearsed hundreds of times, never once stepping on a single piece. The fabrics flutter subtly as he passes, responding to the smallest shifts of air.*
> *The music intensifies briefly, then softens again. The dancer lowers himself slowly, crouching until his body nearly touches the ground. As he rises, a drone ascends with him, lifting in synchrony. He moves again, sometimes fast, sometimes slow. The drone follows, adjusting its position with a gentle delay, as if tethered to him by an invisible thread.*
> *Trailing behind, the drone's airflow causes the fabric to rise and fall, forming transient ridges and swells. Pockets of air become trapped beneath the panels, lifting gently as the drone passes and settling when it moves away. Around him, the stage begins to become a living environment of an ocean of fabric. Waves folding in and out, as if the dancer has submerged in a sea of motion. Each gesture is extended through the material. The stage amplifies the dancer's force, turning his movements into a palpable field of energy. One that spreading outward from the dancer, through the stage, and into the audience.*



Figure 9: Illustration that depicts the use of drones and its airflow as dance performance. The bottom is consisted of overlapping strokes that simulate layers of wave and ocean, rising with the present of the drone. The misty stroke gives a soft and gentle atmosphere of the movement of both the dancer and the drone.

4.2 Interactive Play Space

This scenario was inspired by a playful moment when my peer and I pushed the flying drone toward each other. We were both immersed in a moment of collective engagement—sharing a common goal and responding to each other through the movement of a shared agent. Building on that moment, this scenario adapts the *Wave from Pushing* interaction and the application on shimmering panels, imagining a shared play-space where drones become mediators of connection (Fig. 10):

> *Sally steps into the room. Darkness settles around her, but not completely. As her eyes adjust, she begins to see more clearly. In the center of the space, a low, square-shaped shimmering plane stretches out before her. Reflecting the lights, it creates soft blue reflections across the walls and ceiling.*
> *At each of the four corners stands a platform, slightly raised, like islands in a glowing sea. On three of them, people are already standing. One platform remains empty. She makes her way toward it and steps up. Then she notices the drones. They hover between the platforms, moving slowly. One drifts toward her. She watches, unsure of what to do. Instinctively, she raises her hand and makes a gentle pushing motion through the air.*
> *The drone responds, floating backward in a smooth, wave-like trajectory toward another platform. As it passes above the shimmering plane, its airflow stirs the reflective pieces beneath, leaving a visible trail. The reflective fragments flutter and lift gently, disturbing the once-calm surface and causing the soft*

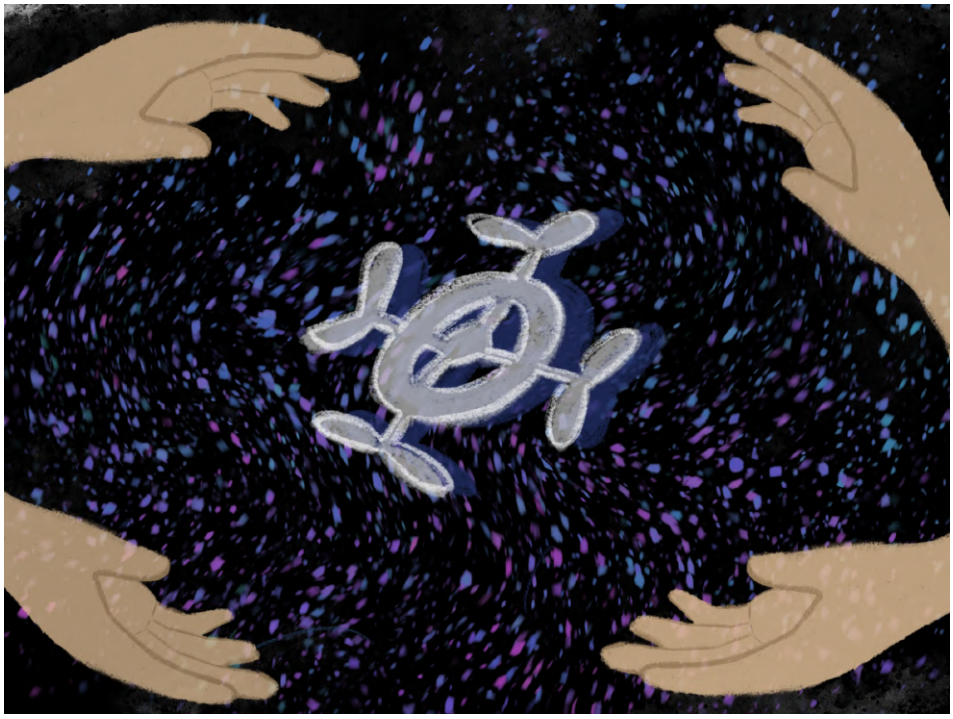


Figure 10: Illustration that depicts the use of drones and its airflow in interactive play space. The glimmers in the background represents the reflecting lights of the shimmering pieces. The glimmers also slightly twisted, showing that the stillness and could be disturbed by the drone. Four hands in the corners give a hint to the interactive nature of this scenario.

> *blue reflections around the room to tremble. The effect moves outward from her, like a ripple of a wave expanding across the surface, resembling an underwater dreamscape.*
> *Across the room, another person lifts their hand in response, catching the drone and sending it to another platform. It is like playing catch, but slower, more deliberate, and poetic. At that moment, Sally feels it isn't just about controlling drones. It is about making connections with the others in the room and the space they share through the drone.*

5 DISCUSSION

This thesis presents an exploratory design process grounded in a Research-through-Design (RtD) approach [47], focusing on movement-based human-drone interaction within a somaesthetic worldview [32]. The work unfolded as a series of iterative explorations, beginning with the lived experience of freediving and culminating in a set of experiential concepts. Within the RtD framework, design artifacts are manifestations of the designer's active choices, with each decision embodying their intentions, values, and beliefs [9]. Knowledge is generated throughout the design process — the ongoing reflection, iteration, and engagement with materials [33]. This positions design as a meaning-making practice, where meaning emerges and evolves through the act of exploring, interpreting, and making.

In this context, I begin by discussing how meaning was constructed through the design process presented in this thesis, where its form is fluid, unstable, and constantly changing. This discussion is organized into five parts: first, the continuous translation between abstract and concrete elements; and second, the navigation between and within

different design mediums. I then explore how I somatically engaged with the process, forming meaningful bonds with myself and drones. Following this, I reflect on the use of a first-person micro-phenomenological methodology, and finally, I address the limitations of this thesis.

5.1 Translation between Concrete and Abstract

The exploratory process described in this thesis reveals a continuous and reciprocal translation between concrete and abstract elements. In this context, concrete refers not only to what is physically perceivable, such as bodily movement or drone behavior, but also to expressive forms that support understanding and discovery. These include texts, sketches, and analogies drawn from real-world phenomena. In contrast, abstract refers to conceptual or interpretive elements, such as experiential qualities or design metaphors that emerge from engaging with concrete experiences.

At a broader level, the thesis traces a design thread that begins with a tangible, lived activity, freediving, and moves toward the extraction of an abstract movement quality, force projection. This quality was treated as a generative metaphor that informed subsequent design ideation. From there, the exploration extended into the realm of physical drone interaction and aesthetic experience, where this abstract quality was re-materialized through interactive expressions. A similar pattern of transitioning between concrete and abstract can be seen in previous somaesthetic design work. For example, Drone Chi combined drone materiality and movement with the abstract somaesthetic qualities of Tai Chi, slow, focused, and smooth movement [19].

Here, I reflect on the methodological tactics I employed to support these transitions. First, visualization techniques, such as the force sketches, translated the imagined sensation of force while performing bodily movements into spatial indicators such as muscular intensity and directional orientation. In addition, I frequently relied on descriptive analogy to express wordless experiential feelings [12] through vocabularies that support understanding and communication [22]. For example, I described a visual-somatic sensation as ‘visual version white noise’ and the drone’s push-away movement as ‘repelling magnets’. Another key tatic was the use of extraction techniques, which enabled the detachment of a concept from original context and transfer into another design medium. For instance, the analysis of micro-phenomenological reflections led to the formation of the experiential quality force projection, while movement analysis gave rise to the design metaphor waves. These concepts were then used to inform design of both drone interactions and aesthetic explorations. In the bigger picture, such techniques are not unique to this thesis. In the field of somaesthetic design, similar techniques are commonly employed to externalize and reflect upon bodily experience. Tools such as the body sheet or body map [1, 13] support the act of translating internal somatic awareness into external representations that allow communication.

Importantly, this translation between concrete and abstract was neither linear nor tidy. Throughout the process, concrete and abstract elements were deeply intertwined. They often coexist simultaneously, with blurred boundaries between them. Moreover, smaller acts of translation were embedded within larger processes, revealing this type of translation is layered and recursive. For example, while exploring the *Follow like Fins* interaction, the embodied experience of engaging with the drone was articulated through an analogy to a real-world phenomenon: freediving fins. This analogy helped express the sensation of the connectedness to an extension of the body. At the same time, fins also became an abstract design metaphor, *Follow like Fins*, which guided the creation of an interaction pattern incorporating delayed drone responses and could more closely resemble the original bodily sensation of freediving. Through these intertwined layers of concrete and abstract, the design trajectory gradually unfolded.

This translation process can be understood as a continuous practice of reframing, a central activity in design thinking, as described by Dorst [5]. In this view, meaning arise from the designer’s ability to reinterpret situations and construct new frames that guide exploration. In this thesis, such reframing occurred through shifts between concrete expressions (bodily movement, sketching, and material engagement) and abstract constructs (design metaphors and experiential

qualities). Each act of translation invited reinterpretation and provided a renewed sense of what was meaningful in the design phase. For designers engaging with similar processes, embracing the messiness and ambiguity of translation, rather than seeking a linear path from abstract ideas to concrete prototypes, can open up opportunities for richer meaning-making and deeper somatic insight.

5.2 Navigation between and within Design Mediums

Another important dimension of meaning-making lies in the navigation across different design mediums. Design medium here refers to the specific vehicle where concepts could be articulated, explored or realized. A medium may include physical materials (e.g., drones, inflatables, shimmering panels), phenomena (e.g., airflow), embodied actions (e.g., bodily movement), or representational forms (e.g., sketches). Moreover, each medium possesses its distinct expressive possibilities. This thesis therefore unfolded through a series of explorations and conversation across the possibilities of different mediums.

Throughout the design process, connections were drawn between seemingly unrelated elements by identifying shared or complementary expressive possibilities, ultimately forming a coherent experiential thread. For example, drones possess an expressive quality through the air displacement generated by their propellers. Shimmering panels, in turn, responded to this expressive quality and reflected its influence through their own expressive channel. On their own, the panels' fluttering might appear meaningless. However, when placed in relation to the drone's airflow, it became a visual representation of waves in a freediving-inspired, somaesthetic drone interaction. Similarly, the drone and the soma share a common expressive channel—dynamic movement—which can be bridged through the metaphor of force projection. These examples reveal an axis of the design's meaning-making process, aligning with Schön's view of design as a reflective conversation with materials, the design situation [33], and the design worldview [32]. From this perspective, meaning, understanding and intention are not predetermined but emerge and evolve through ongoing engagement with the design process.

On the other hand, while each medium came with its own expressive possibilities, it also came with its limitations. This became especially evident when designing with a technological agent. While designing with drones, I constantly experienced technological failures and limited capability. Rather than treating these breakdowns as problems to be solved through a human-centered perspective, I came to understand them as part of the drone's character and treated them as materials to co-evolve with [6]. For instance, the limited sensing range and the difficulty of achieving delicate movement led me to reconstruct the drone's behavior into simple yet expressive forms, rather than striving for seamless responsiveness. As the process progressed, I cultivated a deeper understanding of the drone, such as learning to recognize its status and identify causes of failure. Moreover, I adapted my own behavior to its constraints, such as avoiding the blockage of Lighthouse signals with my body or keeping the drone within its tracking area. These adaptations were a mutual shaping of behaviors [36], implying a broader relational understanding of the drone as a partner. (A more detailed discussion of this can be found in Section 5.3 Somatic Engagement.)

This reflects the practice of designing-with, as proposed by Ron Wakkary, which frames the human designer and nonhuman agent as part of a shared assembly, each contributing to the shaping of the design outcome [45]. Within this view, technological constraints are not merely obstacles to be overcome, but opportunities for designers to engage more deeply with the material. By embracing both the possibilities and limitations of a design medium, designers can approach the process as an open-ended negotiation, one that invites the co-shaping of meanings through attentive engagement with their materials.

5.3 Somatic Engagement

Somatic engagement is a process of establishing a reflective and participatory dialogue with one's own soma. It involves tuning into, interpreting, and responding to embodied sensations that are often tacit, multilayered, and situated within specific contexts of interaction. In this thesis, somatic engagement unfolded across multiple layers, including reflecting on bodily experience, interacting with a responsive agent, and tuning into the agent's behavior. Together, these layers shaped how I perceive myself, my experiences, and the relationship with my design companion – drones.

The first layer of somatic engagement unfolded during the micro-phenomenological process, where I turned inward to reflect on the lived experience of freediving. Re-engaging with the activity allowed me to articulate the nuances that I could not express at the time of the original encounter. The experience itself previously felt like tangled threads, something I knew was rich and layered, yet I could not clearly see or separate its contents. Through M-P methodology, these threads, the implicit sensations, were slowly drawn out and structured into clearer, more coherent vocabularies. For instance, micro-events that had previously gone unnoticed became visible through the guided recollection. One such detail was the rhythm of bodily relaxation. The reflection revealed how this process moved sequentially from the body's core to its extremities, in a flow I hadn't consciously perceived before. Moreover, the method also heightened my awareness of the surrounding environment and its mutual influence with my soma. Initially, I merely recognized that navigating underwater felt unfamiliar and difficult. However through M-P reflection, I realized this discomfort stemmed from my body's habitual reliance on land-based, two-dimensional movement. Postures that felt correct on land, such as lifting the head or opening the chest, disrupted vertical control in the water's three-dimensional space. These rich details reconfigured how I remembered and valued the experience, turning what was once tacit and intuitive into a consciously accessible resource for reflection and design thinking. This reconfiguration did not remain in the memory alone but also informed my subsequent diving experiences. When I returned to the water this summer, I possessed a heightened somatic awareness. To be specific, I consciously adjusted my posture in real time, actively avoided the instinctive land-based posture of lifting my upper body. As the result, I found myself navigating the underwater space with greater ease. These adaptations were embodied responses grounded in a deeper understanding of how my movements and actions interacted with the environment. This somatic awareness allowed me to move more fluidly and purposefully, no longer working against the water but instead moving with it. In this way, the micro-phenomenological process served both as a retrospective tool and as a means of enhancing future somatic engagement and embodied action.

The second layer of somatic engagement took shape during the embodied interactions with drones, where over time, the drone began to feel less like an alien, external object and more like an extension of my body. Through patterns of interaction, drones extended both my perceptive and expressive capabilities, becoming a medium which I could sense through and attune to. This was evident in both interaction concepts developed in this thesis. In *Wave from Pushing*, the force applied by the body was externalized through the drone's responsive movement, allowing me to perceive and sense my force through the drone as an indicator. In *Follow like Fins*, I synchronize my movements in order to form a rhythmic attunement between the soma and drone, creating a somatic interaction loop. These adaptations between our bodily selves with the extension aligns with Merleau-Ponty's description of the relationship between a blind person and their stick, "the stick is no longer an object perceived by the blind man, but an instrument with which he perceives. It is a bodily auxiliary, an extension of the bodily synthesis. [25]" Throughout this thesis, similar assemblages emerged: between stick and blind user, drone and human, fin and diver. These extensions become part of a unified perceptual and reacting system. Svanæs also explores this dynamic in his work on integrating a mechanical tail with his own body, indicating the reshape of the body's boundaries is possible [42]. What emerged through these assemblages is a form of somatic awareness not only of the self but of a hybrid body, where human and extension form a sensing, reacting

whole. This engagement of a hybrid body therefore invited a subtle dialogue to uncover the entangled relationship between bodies and extensions.

The third layer of somatic engagement emerged as I gradually attuned to the drone's behavior over time. This attunement involved developing a bodily sensitivity to the drone's status and constraints, especially in moments of failure. Initially, failures were approached through a routine debug process, established as part of my growing experience base with drone interaction: checking the battery level, inspecting propellers, diagnosing signal or connection loss, and restarting the system before identifying potential coding issues. Gradually, I began to recognize specific patterns of failure intuitively: a drone slowly drifting off or sudden vertical drop pointed to battery depletion, acceleration in drifting away suggested signal issues, and an uncontrolled burst at takeoff indicated problems of propellers. These interpretations were no longer based purely on analytical reasoning but grounded in embodied familiarity, as if my soma had learned to 'read' the drone's signals. Over time, this sensitivity extended beyond interpretation to adaptation. I adjusted my positioning to avoid blocking Lighthouse signals and modulated expectations in response to the drone's limitations. Rather than treating the drone as an imperfect technology to be optimized, I began to regard it as a fragile design companion, one that demanded attentiveness and care. In this evolving relationship, I was not merely a designer but also a listener.

Although this work did not explicitly draw on the concept of mechanical sympathy, similar qualities emerged throughout the design process. In terms of *machine-agency*, the drones shape my behavior through their constraints and expressive capacities; *oscillations* arose through the reciprocal adjustments between my bodily actions and the drone's responses; and the result interaction revealed an *aesthetics pursuit* where the decision was guided by ongoing exploration and negotiation of what felt meaningful in the context of embodied interaction [20]. These qualities reflect the unfolding of somatic engagement, where perception, action, and reflection are co-shaped through the embodied dialog between human and drone.

5.4 Micro-phenomenological Methodology through First-person Perspective

In this thesis, I expanded the application of micro-phenomenological (M-P) methodology to a new setting — a reflexive first-person somaesthetic self-exmination. The process articulated the somatic richness of my past freediving experience, which unfolded in two parts. First, I conducted a micro-phenomenological interview. However, unlike conventional M-P interviews, organized through a second-person perspective, where the researcher interviews an external participant to collect data [30], this thesis adopted a first-person adaptation. In this case, I was interviewed by a facilitator experienced in M-P inquiry to elicit detailed descriptions of my own lived experience as a researcher. Second, I employed a method similar to what Sparby describes as micro-phenomenological self-inquiry [37]. In this session, I construct an autobiography, which is a common method adopted in discovering and reflecting on somatic movement quality, to document experiential data from the previous session. This was further enriched by reviewing past freediving videos and conducting a self-guided micro-phenomenological elicitation, which together provided additional details and depth to the experience.

This hybrid method allowed the recall of subtle nuances by re-immersion into multiple layers of past experiences. It is particularly suitable for researchers with limited experience in micro-phenomenological inquiry. The first stage, guided by a professional interviewer, not only provided the foundation for the multi-dimensionality of the experience but also allowed me to become familiar with the structure of micro-phenomenological dialogue. In the second stage, freed from the time constraints of an external interview, I could engage with the experience in a more relaxed and self-paced manner. Under this setting, a follow-up micro-phenomenological self-inquiry enabled me to delve into a deeper layer of the experience, revealing additional details of somatic sensation.

Current soma design methods emphasize the importance of 'attending to the soma' [14], encouraging designers to become attuned to their bodily sensations during the design process. Moreover, techniques such as autoethnography [12] and embodied sketching [24] are commonly used to surface and reflect on these embodied experiences. However, while these methods are effective for capturing sensations as they arise, challenges may present when attempting to articulate the fine-grain somatic dimensions of past experiences. Therefore, I suggest that M-P methods can serve as a valuable complement to conventional soma design practices, particularly when designers seek to retrieve and describe embodied experiences retrospectively. Moreover, this approach is accessible even to researchers inexperienced in micro-phenomenological inquiry, as they can begin by participating as interviewees before attempting self-inquiry. Taken together, this thesis demonstrated an approach that expanded the methodological toolkit of soma design, offering a deeper way to access and articulate past embodied experiences.

5.5 Limitations

This thesis intentionally adopts a first-person perspective to access and interpret the designer's own embodied experience throughout the design process. This approach aligns with established practices in soma design, which prioritize the richness and dimensionality of subjective, felt experiences that are often difficult to articulate through third-person methods. However, at the same time, I acknowledge that this methodological choice centers my own embodied, cultural, and aesthetic worldview. It places me in a powerful interpretive role, where meaning is shaped through my lens and internal reflection. While the work also explores how the drone shaped my behavior and interaction patterns, I remained the one making design decisions and constructing the narratives. Moreover, the introspective nature of this process further amplifies that asymmetry. The absence of external perspectives means the knowledge generated is inherently situated in my experience, offering depth but also limiting the diversity of interpretations.

Extending this stance, the assessment of the effectiveness and usefulness of the first-person micro-phenomenological (M-P) interview for accessing subtle, embodied details of past experience relies solely on my own account. M-P might not be as generative in all research contexts, as many factors influence the effectiveness of the method, such as how far in the past that an experience took place. As this was a first attempt, the usefulness of this adaptation remains self-reported, and the method itself still requires additional trials before it can be considered a reliable tool within design contexts.

On the other hand, while this work adopts storytelling to bridge the gap between physical prototypes and conceptual insights, it does not reject the value of physical prototyping. Although storytelling serves as a tool to expand imagination and discover potential values, physical engagement remains essential for accessing tactile feedback and somatic nuance. As such, relying solely on narrative may risk losing the embodied immediacy and material resistance that are often central to somaesthetic interaction.

6 CONCLUSION

This thesis set out to explore how movement qualities inspired by freediving could inform the design of somaesthetic human–drone interaction. Through embracing a first-person Research-through-Design approach, the lived experience of freediving was revisited using micro-phenomenological methods, allowing for the extraction of subtle somatic insights. These insights, notably the experiential quality of *force projection*, were translated into interaction patterns of drone, *Follow like Fins* and *Wave from Pushing*, and further augmented the aesthetic experience through physical materials. The work culminated in speculative scenarios that extended these insights into immersive experiences.

The process revealed that meaning in somaesthetic interaction does not reside in the design outcome alone, but emerges through the entangled relationship between soma, material, and technology. Freediving, in this context, was not merely a thematic inspiration but a somatic lens that shaped how interaction was perceived, enacted, and

understood. The drone revealed its expressive potential when approached as a collaborator in movement design, capable of co-shaping the interaction through its responsiveness and constraints. The dialogue between these elements unfolded through continuous translation between concept and realization, and across different materials and media. This work contributes to the soma design discourse by demonstrating how personal, embodied experience can serve as fertile ground for design ideation, and how sustained somatic engagement throughout the process can deepen the relational and interpretive dimensions of interaction.

ACKNOWLEDGMENTS

I would like to express my gratitude to my supervisor Rachael Garrett for the invaluable assistance and guidance throughout my research. throughout this work. I am also deeply thankful to Joseph La Delfa for the technical support with the drones and for sharing inspiring ideas that shaped the direction of this work. My appreciation goes as well to Katerina Stepanova for conducting the micro-phenomenological interview with me. Finally, I want to thank my partner, friends, and family for showing encouragement and support along the way.

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